

API Protocol: Schedule



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API PROTOCOL SCHEDULINGCFG

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Overview:

This API can be used for the scheduling configuration in a Duotecno node.

The base url for this API is: **[IP-Address]:8080/**

[IP-Address]: The IP-Address of the Duotecno node.
The port is always 8080.

Possibilities:

- Request tag, time and date of last changes.
- Full scheduling configuration: Upload / Download / Erase / Request summary.
- Single schedule item configuration: Request / Edit / Erase / Delete
- Single schedule item set state on/off
- Alarm clock: Request / Set / Delete
- Alarm clock: set state on/off

Not every node supports all these API functions.

The configuration capabilities can be requested via the API call **/nodecfginfo.json** (See general documentation API)

The alarm clock items and events are stored in the same scheduling database but they are accessed by different API functions. When requesting or setting scheduling items the alarm clock is not addressed. When setting the alarm clock the other scheduling items are not affected.

The virtual unit assigned to the alarm clock must be unique to the alarm clock. It cannot be used in the normal scheduling.

When uploading scheduling configuration where a virtual unit is used and already assigned to the alarm clock there will be an error returned.

When assigning a virtual to the alarm clock which is already used in the scheduling there will also be an error returned.

NOTE:

When the tag has changed FIRST download the current configuration of the scheduling(s). Edit the needed values and upload the scheduling(s) again.

NOTE:

When the Clock alarm settings changes via this API or via the touchscreen user interface the tag will be renewed.

Endpoints - Overview

HTTP Method	URL	Description
GET	/schedulecfg/tag	Request tag last changes
GET	/schedulecfg/info	Request summary.
GET	/schedulecfg/[nodeAddress]/[unitAddress]	Single item: Request
POST	/schedulecfg	Single item: Adjust/erase
POST	/schedulecfg/[nodeAddress]/[unitAddress]/state	Single item: Set state on/off
DELETE	/schedulecfg/[nodeAddress]/[unitAddress]	Single item: Delete
GET	/schedulecfg/clockalarm	Alarm clock: Request
POST	/schedulecfg/clockalarm	Alarm clock: Set
POST	/schedulecfg/clockalarm/state	Alarm clock: Set state on/off
DELETE	/schedulecfg/clockalarm	Alarm clock: Delete
GET	/config/download/scheduling.json	Download configuration
POST	/config/upload/scheduling.json	Upload/erase configuration

Json data objects - Overview

Json Data object – Schedule Item.

This object represents a schedule item. This item is assigned to a LCD virtual unit or a sensor unit. A schedule item can hold one or more schedule events. When the time regime, day and time are equal the unit executes the time event.

Key	Value	Description
description	String	Read/Write Optional: A text describing the function of the schedule item. This text should be displayed to the end-user and gives information about the scheduling item. When there is no text description this field is omitted.
empty	Boolean	Read only This value indicates if there are scheduling events. This property is meaningful when a summary of the scheduling is requested. This key has no meaning when editing or uploading data.
events	Array	Read/Write Array of scheduling events to be executed by this unit.
name	String	Read only The name of the unit assigned to the scheduling.
nodeAddress	Integer	Read/Write The node address of the unit belonging to this scheduling item.
unitAddress	Integer	Read/Write The unit address of the unit belonging to this scheduling item.
stateOnOff	boolean	Read/Write This property is only needed for a schedule item of type sensor. When “true” the scheduling is active, when “false” the scheduling is not active.

		(A Schedule item for a virtual is always active unless it has no events).
type	String	Read/Write The value indicates which type of unit is connected and hence which actions will be executed. Possible values are: "sensor" : Sensor unit. "virtual" : Virtual unit. "unknown" : Only returned when requesting info: the unit was not found in the database is unknown and it's events will not be executed.
userAdjustable	Boolean	Read/Write This value indicates if the end-user can change this scheduling item. Items that the end-user cannot change should not be displayed in the end-user configuration.

Json Data object – Schedule event.

This object holds the time when to be executed and the action to be executed by the schedule item.

Key	Value	Description
action	String	Read/Write The action to be executed. The action depends on the type of the schedule item. Possible values see tables below.
week	String	Read/Write The scheduling has four different week types. This value holds the week in which the event is active. Possible values: "week1", "week2", "week3", "holiday"
day	String	Read/Write Day of the week. Possible values: "mon", "tue", "wed", "thu", "fri", "sat", "sun"
time	String	Read/Write The time when the event has to be executed. Syntax: "hh:mm" – 2 digits - 24h mode. Note: Events for temp sensor are always in intervals of 15 minutes. Note: Events for virtuals are always in intervals of 5 minutes.

actions for "sensor":

Value	Description
"sun"	Set sensor on + preset sun
"half_sun"	Set sensor on + preset half sun
"moon"	Set sensor on + preset moon
"half_moon"	Set sensor on + preset half moon
"off"	Set sensor off

Attention: Sensor actions are only executed when the events are interpolated.

Example: 6u00 Sun / 7u45 Moon / 8u00 Sun (This is ok.)

Example: 6u00 Sun / 8u00 sun (Not ok – Event 8u will not be activated.)

actions for "virtual":

Value	Description
"puls"	Short puls

"pton"	Short puls toggle – state on
"ptoff"	Short puls toggle – state off
"ton"	Long toggle – state on
"toff"	Long toggle – state off

actions for "unknown":

When an attached unit is not found in the node database it's actions will not be executed.
This value can only be read. It cannot be used to upload configuration.

Value	Description
"unknown"	Unknown action – Will not be executed.

Json Data object – items

The items object is used for full configuration.

Key	Value	Description
items	Array	Array of scheduling item objects.

Json Data object – items_info

The items_info object is used when a summary of the configuration is requested.

Key	Value	Description
items_info	Array	Array of scheduling item objects.

Json Data object – Alarm clock

This object holds the properties of the alarm clock. This item can be attached to a LCD virtual.

Key	Value	Description
day	String	Read/Write Day of the week. Possible values: "mon", "tue", "wed", "thu", "fri", "sat", "sun" This value is only present / needed when the types are "once", and "weekly".
holidayActive	Boolean	Read/Write The alarm clock is valid in the weekregime holiday. "true": alarm clock is active in all week regimes including holiday. "false": alarm clock is only active in week regimes: week1, week2 & week3, not in holiday.
name	String	Read The name of the virtual unit assigned to the alarm clock. When there is no unit assigned this value is omitted.
nodeAddress	Integer	Read/Write The node address of the virtual unit that is assigned to the alarm clock. This virtual will be executed when the clock event is fired. When read and this value is 255 then there is no virtual assigned.

		The value 255 cannot be written. (use DELETE to remove the virtual)
unitAddress	Integer	Read/Write The unit address of the virtual unit that is assigned to the alarm clock. This virtual will be executed when the clock event is fired. When read and this value is 255 then there is no virtual assigned. The value 255 cannot be written. (use DELETE to remove the virtual)
stateOnOff	Boolean	Read/Write Alarm clock on/off. "true": Alarm clock on "false": Alarm clock off
time	String	Read/Write The time when the event has to be executed. Syntax: "hh:mm" – 2 digits - 24h mode.
type	String	Read/Write This value defines how the alarm clock will act. Possible values see table below.

Alarm clock: different types.

Value	Description
"once"	The alarm clock event will be fired once at the specified time and week day. After the event is fired the state of the alarm clock will be set to off.
"weekly"	The alarm clock event will be fired weekly repeated at the specified time and week day.
"daily"	The alarm clock event will be fired daily repeated at the specified time and week day.
"weekdays"	The alarm clock event will be fired weekdays repeated at the specified time and week day.

Json Data object – State on/off

This object is used to change the on/off state of a schedule item or the clock alarm.

Key	Value	Description
stateOnOff	boolean	Set the scheduling active for a schedule item. "true" is Scheduling active, "false" is disable scheduling.

Json data object – database tag info

This info holds a tag, date and time when the last changes were made.

Key	Value	Description
date	string	Optional: The date of last changes. "dd.mm.yyyy"
time	String	Optional: The time of last changes. "hh:mm:ss".
tag	String	A 20 char tag. The first time requested or when there is no configuration (after a factory init) this can be an empty string.

Json Data object – Status message

Json responses are returned with http status code 200. They are a response to a POST/PUT http message that holds a Json data object in the http body.

Json responses always hold a “status” key.

When the value is “ok” then the http request was a success.

```
{  
  "status": "ok"  
}
```

When the value is “error” then an error happened and optional the “error” and “error_detail” key are added:

```
{  
  "error": "invalid_request",  
  "error_detail": "Invalid refresh token",  
  "status": "error"  
}
```

Example: When a node doesn't support the configuration it will give an error response:

```
{  
  "error": "Scheduling config failed",  
  "errorMsg": "Scheduling is not allowed for this node - not the master node.",  
  "status": "failed"  
}
```

Units not found in the node database:

Scheduling events are only invoked when the attached unit is available in the node database.
Therefore:

When requesting info:

When the unit is not present the json item object "type" field will be "unknown" and the json event object "action" field will also be "unknown".

Example:

```
{
  "empty": false,
  "events": [
    {
      "action": "unknown",
      "day": "sun",
      "time": "05:30",
      "week": "week1"
    }
  ],
  "name": "",
  "nodeAddress": 100,
  "stateOnOff": false,
  "type": "unknown",
  "unitAddress": 0,
  "userAdjustable": false
}
```

Possible the client application can notify the user these actions will not be invoked.

When uploading configuration:

It is not possible to upload events with these fields "unknown". The client application must know there type.

Though it is possible to upload events for units that are (at the moment) not present in the node database. Later when they are added via pnp or browse for nodes the events will be executed.

The alarm clock:

The alarm clock can only be attached to a virtual unit that is part of the node.

When uploading configuration for a unit that is not present in the node or a wrong unit type the configuration will fail and an error will be returned.

Request tag, time and date last changes

When the configuration is changed the time, date and a tag are generated and stored.

A new tag is also generated when:

- The state On/Off changes from a scheduling item.
- The Alarm clock settings changes. (via API or touchscreen user interface).

These values can be requested by the API.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	GET
Endpoint	/schedulescfg/tag
Response	Status code 200 + Json database taginfo object

Response:

```
{
  "date": "09.11.2020",
  "tag": "BGFZR7J8p7rKDCeZQlf6",
  "time": "11:47:01"
}
```

The response can be empty tag value.

```
{
  "tag" : ""
}
```

This is possible when:

- There is no scheduling configuration present.
- Configuration has not been changed since a firmware update supporting this API.

Download the full configuration

Download the full scheduling configuration from the node.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	GET
Endpoint	/config/download/scheduling.json
Response	Status code 200 + Json moods data object

Response:

HTTP Status code 200 + Json schedule data object:

```
{
  "items": [
    {
      "description": "Deze scheduling kan niet door de klant aangepast worden.",
      "empty": false,
      "events": [
        {
          "action": "sun",
          "day": "mon",
          "time": "06:00",
          "week": "week1"
        },
        {
          "action": "moon",
          "day": "mon",
          "time": "07:00",
          "week": "week1"
        }
      ],
      "name": "Living",
      "nodeAddress": 7,
      "stateOnOff": true,
      "type": "sensor",
      "unitAddress": 96,
      "userAdjustable": false
    }
  ]
}
```

Upload the full configuration:

Upload the full scheduling configuration to the node.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/config/upload/scheduling.json
Data body	Json moods data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
  "items": [
    {
      "description": "Deze scheduling kan niet door de klant aangepast worden.",
      "events": [
        {
          "action": "sun",
          "day": "mon",
          "time": "06:00",
          "week": "week1"
        },
        {
          "action": "moon",
          "day": "mon",
          "time": "07:00",
          "week": "week1"
        }
      ],
      "nodeAddress": 7,
      "stateOnOff": true,
      "type": "sensor",
      "unitAddress": 96,
      "userAdjustable": false
    }
  ]
}
```

Response on success:

```
{
  "status": "ok"
}
```

Response on failure : An item is attached to the clockalarm.

```
{
  "error": "Database error",
  "errorMsg": "Schedule item with address 252,4 is already assigned to the alarm clock.",
  "status": "failed"
}
```

Erase the full configuration:

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/config/upload/scheduling.json
Data body	Json moods data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
  "items" = [ ]
}
```

Response on success:

```
{
  "status": "ok"
}
```

Request scheduling configuration summary.

This request can be used to get a summary of the scheduling items.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	GET
Endpoint	/schedulecfg/info
Response	Status code 200 + Json items_info object

Response:

```
{
  "items_info": [
    {
      "description": "Deze scheduling kan niet door de klant aangepast worden.",
      "empty": false,
      "name": "new U60",
      "nodeAddress": 7,
      "stateOnOff": true,
      "type": "sensor",
      "unitAddress": 96,
      "userAdjustable": false
    }, ... ]
  }
```

Single schedule item: Request

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	GET
Endpoint	/schedulecfg/[NODEADDRESS]/[UNITADDRESS]
Response	Status code 200 + Json mood object

[NODEADDRESS] : Node address of the unit / schedule item

[UNITADDRESS]: Unit address of the unit/schedule item.

Response:

```
{
  "description": "Deze scheduling kan niet door de klant aangepast worden.",
  "empty": false,
  "events": [
    {
      "action": "sun",
      "day": "mon",
      "time": "06:00",
      "week": "week1"
    },
    {
      "action": "moon",
      "day": "mon",
      "time": "07:00",
      "week": "week1"
    }
  ],
  "name": "Living",
  "nodeAddress": 7,
  "stateOnOff": true,
  "type": "sensor",
  "unitAddress": 96,
  "userAdjustable": false
}
```

Single schedule item: Create / edit

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/schedulecfg/
Data body	Json schedule item + events data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
  "description": "Deze scheduling kan niet door de klant aangepast worden.",
  "events": [
    {
      "action": "sun",
      "day": "mon",
      "time": "06:00",
      "week": "week1"
    },
    {
      "action": "moon",
      "day": "mon",
      "time": "07:00",
      "week": "week1"
    }
  ],
  "nodeAddress": 7,
  "stateOnOff": true,
  "type": "sensor",
  "unitAddress": 96,
  "userAdjustable": false
}
```

The fields are the same as when uploading the full configuration.

Response on success:

```
{
  "status": "ok"
}
```

Response on failure: Item already attached to the clockalarm

```
{  
  "error": "Database error: Could not add schedule item",  
  "errorMsg": "Item already attached to the clockalarm.",  
  "status": "failed"  
}
```

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Single schedule item: Erase events

This can be used to erase the events of the schedule item but hold the info (description, user adjustable, ...)

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/schedulecfg/
Data body	Json schedule item data object
Response	Status code 200 + Json status object

Request Json data body:

```
{
  "description": "Deze scheduling kan niet door de klant aangepast worden.",
  "events": [ ],
  "nodeAddress": 7,
  "stateOnOff": true,
  "type": "sensor",
  "unitAddress": 96,
  "userAdjustable": false
}
```

Single Schedule item: Delete

This function deletes the scheduling item.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	DELETE
Endpoint	/schedulecfg/[NODEADDRESS]/[UNITADDRESS]
Response	Status code 200 + Json status object

[NODEADDRESS] : Node address of the unit / schedule item

[UNITADDRESS]: Unit address of the unit/schedule item.

Single Schedule item: Set state on/off

This function can be used when only the state on/off has to be changed.
This is only useful for a sensor unit.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/schedulecfg/[NODEADDRESS]/[UNITADDRESS]/state
Data body	Json schedule on/off item.
Response	Status code 200 + Json status object

[NODEADDRESS] : Node address of the unit / schedule item

[UNITADDRESS]: Unit address of the unit/schedule item.

Request Json data body:

```
{  
  "stateOnOff" : false  
}
```

Response on success:

```
{  
  "status": "ok"  
}
```

Response on failure:

```
{  
  "error": "Database error",  
  "errorMsg": "Failed to set the schedule item state",  
  "status": "failed"  
}
```

Alarm clock: Request

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	GET
Endpoint	/schedulecfg/clockalarm
Response	Status code 200 + Json alarm clock object

Response:

```
{
  "day": "thu",
  "holidayActive": true,
  "name": ",",
  "nodeAddress": 252,
  "stateOnOff": true,
  "time": "06:11",
  "type": "weekly",
  "unitAddress": 4
}
```

Response when the alarm clock is disabled (no virtual unit is assigned):

```
{
  "day": "mon",
  "holidayActive": false,
  "nodeAddress": 255,
  "stateOnOff": false,
  "time": "00:00",
  "type": "once",
  "unitAddress": 255
}
```

Alarm clock: Set

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/schedulecfg/clockalarm
Data body	Json schedule alarm clock item.
Response	Status code 200 + Json status message.

Request Json data body:

```
{
  "holidayActive": true,
  "nodeAddress": 252,
  "stateOnOff": true,
  "time": "06:00",
  "type": "weekdays",
  "unitAddress": 4
}
```

Response on success:

```
{
  "status": "ok"
}
```

Response on failure: Virtual already assigned to a another scheduling item:

```
{
  "error": "Failed the set the clockalarm.",
  "errorMsg": "Cannot assign the virtual to the clockalarm because it is already used as a scheduling item",
  "status": "failed"
}
```

Response on failure: Wrong value for time

```
{
  "error": "Json parser error.",
  "errorMsg": "JSoNCfgScheduleClockAlarm::build failed: key 'time' wrong value for hour 0-24",
  "status": "failed"
}
```

Alarm clock: Set state on/off

This function can be used when only the state on/off has to be changed.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	POST
Endpoint	/schedulecfg/clockalarm/state
Data body	Json schedule on/off item.
Response	Status code 200 + Json status message.

Request Json data body:

```
{
  "stateOnOff" : true
}
```

Response on success:

```
{
  "status": "ok"
}
```

Response on failure:

When the alarm clock is disabled the state cannot be changed and an error message will be returned:

```
{
  "error": "Database error",
  "errorMsg": "Failed to set the clock alarm state",
  "status": "failed"
}
```

Alarm clock: Delete.

This function can be used to remove the assigned virtual to the alarm clock.
When there is no virtual assigned the alarm clock cannot be used.

Supported nodes:

Node	Minimum firmware version
Touchscreen	V66.02.30

Overview:

http method	DELETE
Endpoint	/schedulecfg/clockalarm
Response	Status code 200 + Json status message.

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